

Yining (Carina) Fu

Data Analyst | Statistical Analyst | Biostatistician
Buford, GA | 919-400-0503 | yining2026@gmail.com | LinkedIn | GitHub

Professional Summary

Data analyst with an M.S.P.H. in Biostatistics and experience across operations, public health research, and financial analytics. Proficient in R, SAS, Python, SQL, Tableau, and Excel for data management, statistical analysis, automation, and visualization. Experienced in building performance tracking systems, analyzing workforce and operational data, and translating complex datasets into practical business decisions. Strong background in statistical modeling, machine learning, clinical trial analysis, and cross-functional collaboration.

Education

Emory University

M.S.P.H in Biostatistics

Atlanta, GA

Aug. 2024 – May. 2026

North Carolina State University

B.S. in Statistics, Minor in Economics; Dean's Honors List

Raleigh, NC

Aug. 2019 – Dec. 2023

Experience

GOFO Inc.

Regional Support Specialist

Suwanee, GA

Aug. 2026 – Present

- Built a labor efficiency tracker to monitor workforce productivity across sites in real time, enabling timely staffing adjustments based on operational needs and helping maintain site stability while controlling labor costs.
- Managed contract worker payroll, staffing agency communication, and timesheet records across multiple sites.
- Built a web-based tracking system to monitor staffing agency performance, labor-hour completion, and retention; used turnover data to target training efforts, reducing contract worker turnover by 25% within three months.
- Created an SOP for contract worker pay raises, keeping information consistent across warehouse teams, staffing agencies, and internal departments; reduced employee complaints by 10%.
- Supported the transition from hourly pay to a piece-rate pay model for contract workers, reducing labor costs by an average of \$0.02 per package.

Emory University

Quantitative Research Assistant

Atlanta, GA

Sep. 2025 – May 2026

- Integrated and standardized geospatial, demographic, and public health data across nearly 100,000 records, building dataset documentation and codebooks to support data quality and statistical modeling, enabled a reproducible analysis across the research team.
- Developed Self-Organizing Map (SOM) clustering models using elbow, silhouette, and gap statistic optimization; improved between-cluster variance explained from 0.31% to 24.13%; authored an accompanying manuscript.
- Built Tableau dashboards translating clustering outputs into actionable audience profiles, presenting analytical findings to cross-functional teams to guide intervention planning.

Chinese Academy of Sciences, Institute of Soil Science

Research Assistant

Nanjing, China

Jan. 2024 – Jun. 2024

- Contributed to spatial segmentation and contamination clustering analyses (SAS, R, ArcGIS) to identify heavy metal distribution patterns across farmland regions.
- Preprocessed soil physicochemical and heavy metal datasets using logarithmic transformation and spatial interpolation, generating analysis-ready data for geospatial modeling.
- Interpreted regional soil contamination trends to guide risk assessment and remediation planning, rehabilitating heavy-metal-contaminated farmland by selecting an appropriate soil passivation remediation scheme. Supported evidence-based decision-making for farmland remediation efforts.

North Carolina State University

Undergraduate Research Assistant

Raleigh, NC

Mar. 2023 – May. 2023

- Collected and managed clinical and genetic datasets to investigate gene-cancer associations, supporting study documentation and record-keeping.

- Applied statistical testing (p-value calculation) in Python and generated charts to evaluate treatment effects on mortality outcomes.

Shanxi Securities Company Limited

Financial Analytics

Shanghai, China

Jun. 2022 – Aug. 2022

- Analyzed financial performance and valuation for 500+ public companies using Wind and Choice financial databases, leveraging SQL queries to extract and organize large-scale financial datasets, and collaborated on Python/R automation scripts that reduced manual processing time by 30%.
- Built Excel dashboards and Tableau visualizations that synthesized complex financial data into data-driven recommendations for audiences ranging from peer analysts to senior stakeholders.

Academic Projects

Aedes-Borne Viruses (ABV) Transmission Analysis Project

R

- Contributed to study design and analyzed data from a cluster-randomized, Test-Negative Design trial evaluating indoor residual spraying efficacy for disease control, applying statistical testing to quantify treatment effects across 15- and 30-day windows.
- Built custom R algorithms to detect spatial clustering and high-risk transmission patterns, producing publication-quality tables and figures for intervention resource allocation.
- Built an interactive dashboard and conference poster, and authored a manuscript, to disseminate findings to academic and stakeholder audiences.

Clinical Trials Case Study

SAS

- Managed longitudinal clinical trial datasets using SAS PROC SQL, deriving Kaplan-Meier survival curves, Kappa statistics, and diagnostic accuracy metrics (sensitivity, specificity, PPV, NPV).
- Produced statistical reports and graphical summaries translating diagnostic accuracy results into structured evidence for decision-making.

FAERS Machine Learning Project

R

- Cleaned and merged seven FDA Adverse Event Reporting System datasets to build an analysis-ready dataset for machine learning applications.
- Developed and compared LDA, QDA, and KNN classification models to predict adverse event outcomes, achieving 80% prediction accuracy.
- Evaluated model performance and conducted gap analysis to identify limitations and potential areas for improvement.

Prospective Evaluation of Radial Keratotomy (PERK) Group Project

R / SAS

- Conducted t-tests and ANOVA to assess relationships between patient characteristics and surgical outcomes.
- Produced numerical and graphical summaries to report post-surgical outcome improvements and procedure effectiveness.

Research Presentations

Presented findings from neighborhood risk stratification and transmission dynamics analyses to faculty mentors and public health researchers at Emory University.

Skills

Programming & Data: R, SAS, Python, MySQL, PostgreSQL, Advanced Excel, Data Cleaning, Data Integration, QA/QC, Data Documentation

Statistical Analysis: Regression Analysis, Generalized Linear Models, Hypothesis Testing, A/B Testing, Survival Analysis, Machine Learning, Spatial Statistics

Data Visualization & Reporting: Dashboard Development, Data Visualization, Tableau (Basic), ArcGIS, Publication-Quality Tables and Figures

Research & Analytics: Clinical Trial Analysis, Study Design, Program Evaluation, Test-Negative Design, Workforce Analytics, Labor Efficiency Analysis, Financial Analytics

Languages: Mandarin Chinese (Native), English (Professional)